

Airline Customer Satisfaction – Data Analysis



Mohammad Shah
Abdul Wasay
Faria Akther
Sumandeep Kaur

Group 15

Overview:

- Project Aim
- Our Dataset
 - Pre-processing, and cleaning the dataset for analysis
 - Analytical Methods
- Satisfaction Distribution
- Overall Analysis
- Class based Analysis
 - Eco, Eco Plus, Business
- Recommendations for Airlines
- Conclusion

Project Aim:

- Identify the primary factors that enhance customer satisfaction
- Compare satisfaction factors across class types, and analyze how customer expectations vary among them
- Highlight high impact but low rated service areas



Our Dataset:

- Dataset is from a real airline, with the airline name undisclosed, but the data reflects real passenger experiences (dataset is from Kaggle)
- Exploring the dataset :
 - Dataset shape: (129880 rows, 22 cols).
 - 4 object (categorical) columns (e.g. satisfaction, Customer Type, Type of Travel, Class).
 - 18 integer columns (age, ratings, delays, etc.).
 - 1 float column (Arrival Delay in Minutes after imputation).



Data Preprocessing – Handling Missing Values:

Missing Value Check:

- Used `df.isnull().sum()` to count missing values column-wise.
- Identified missing values in Arrival Delay in Minutes.

Data Preprocessing – Handling Missing Values:

Imputation Strategy:

- Computed the median arrival delay.
- Replaced missing values in Arrival Delay in Minutes with this median.

Why Median Imputation?

- Median is robust to extreme delay values (outliers).
- Avoids dropping rows and losing data.
- Keeps the distribution of delays realistic for analysis.

Analytical Methods:

For Statistical Analysis:

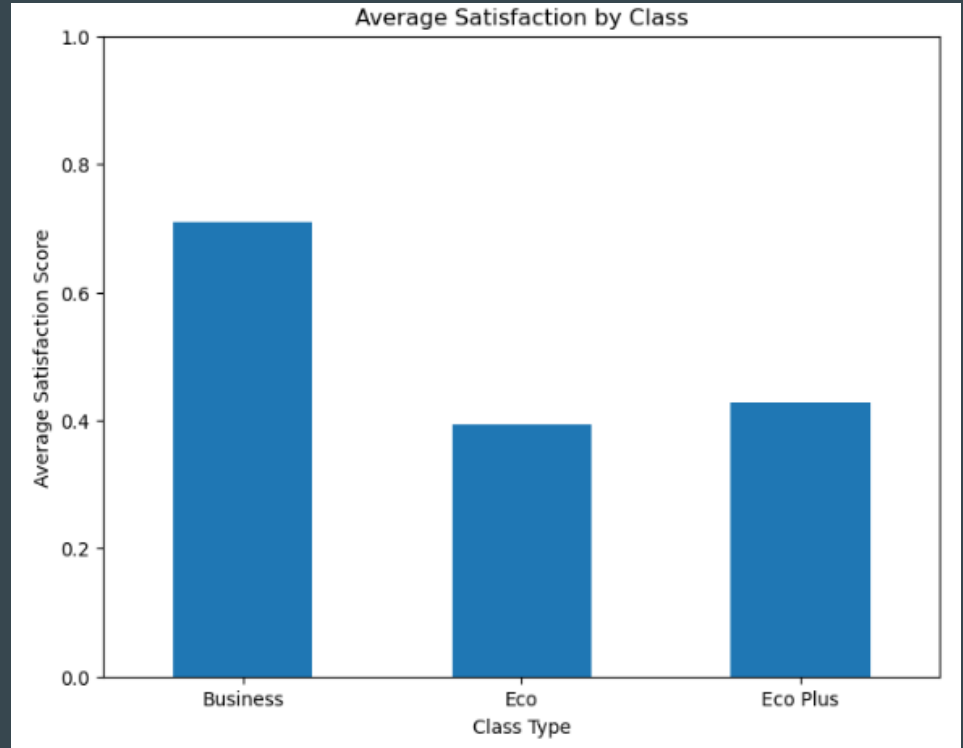
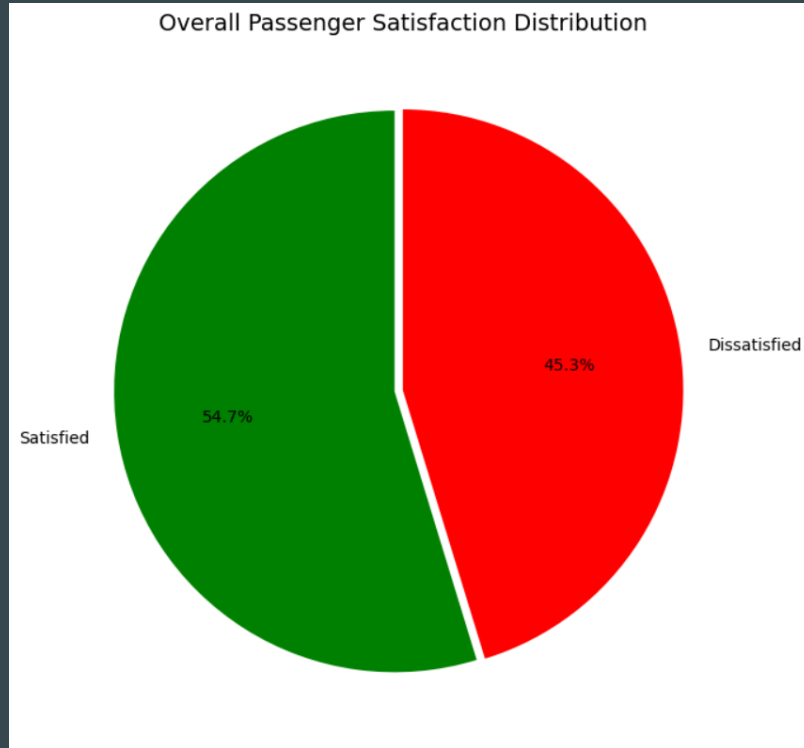
- Started with setting up factors which needs to be considered to derive insights.
 - For eg- seat comfort, inflight entertainment, online boarding, check in services etc are some factors considered.
- Then we setup the correlation of the overall customer experience and the different seating classes with each factor.

Data Visualisation:

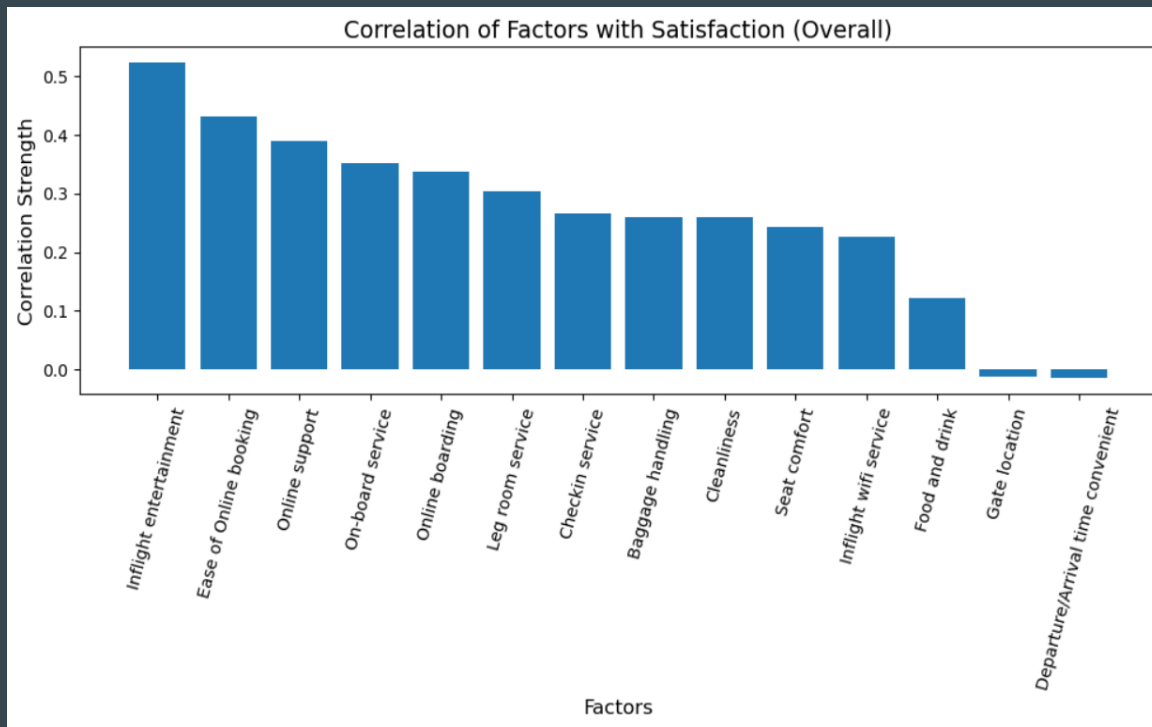
- Bar charts are used to present our findings effectively.

```
Economy Class Shape: (58309, 23)  
Economy-Plus Class Shape: (9411, 23)  
Business Class Shape: (62160, 23)
```

Satisfaction Distribution:

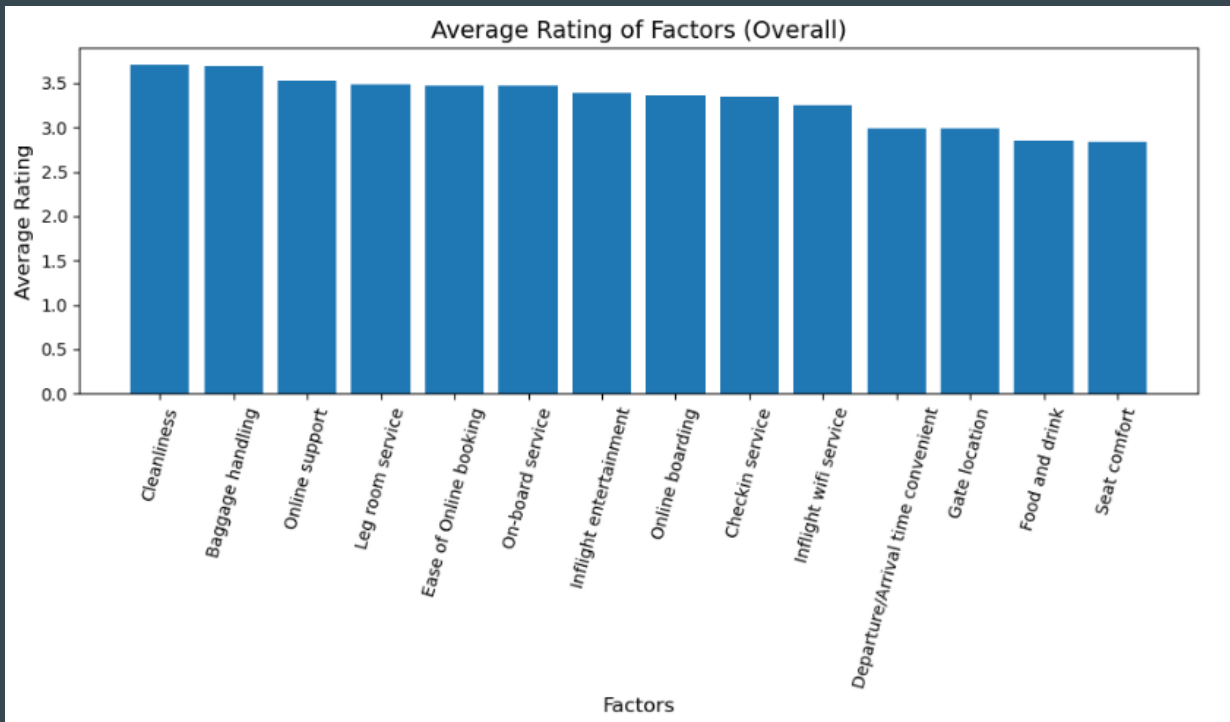


Overall Factor Correlation Analysis:



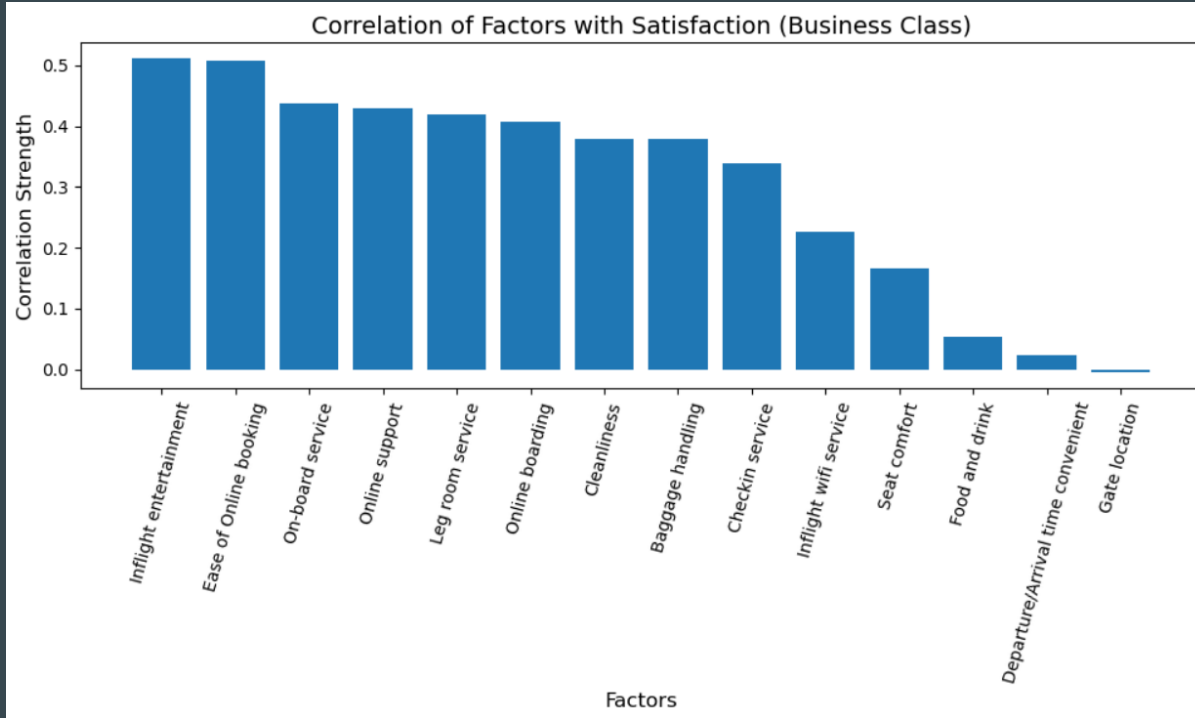
- The ranking highlights two major themes: passengers value digital convenience for the flight, and engaging in-flight experiences during the flight

Overall Average Rating:



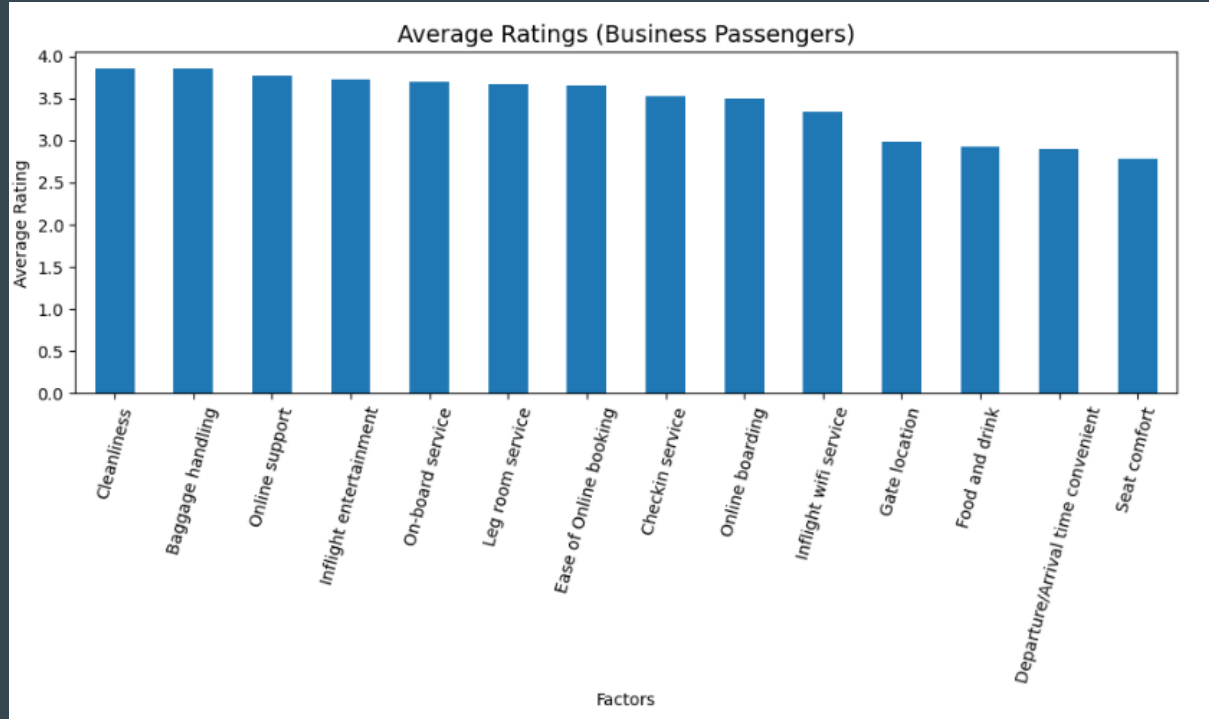
- Shifting from Correlation to average rating shows a new perspective on these factors.
- No factors in this dataset reached an average rating of 4 or above.

Business Class Factor Correlation Analysis:



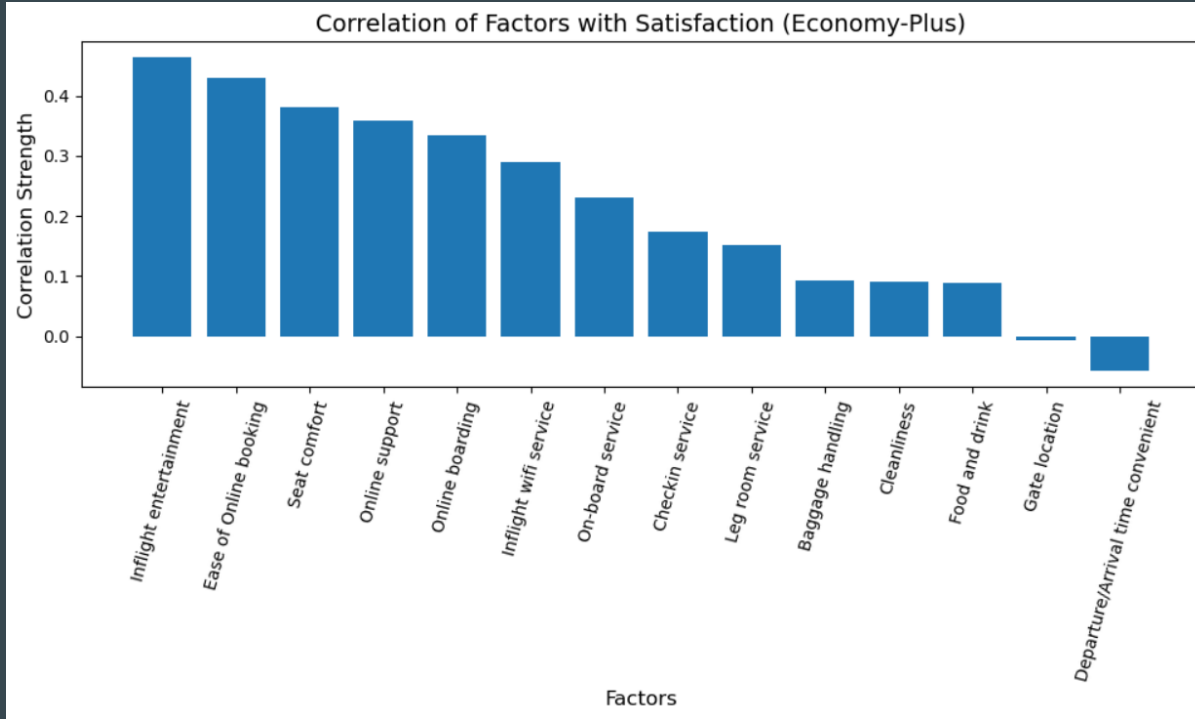
- Based off the rankings in the bar chart, there is high correlation of inflight entertainment, and ease of online booking, meaning a stronger relationship with customer satisfaction.

Business Class Average Rating:



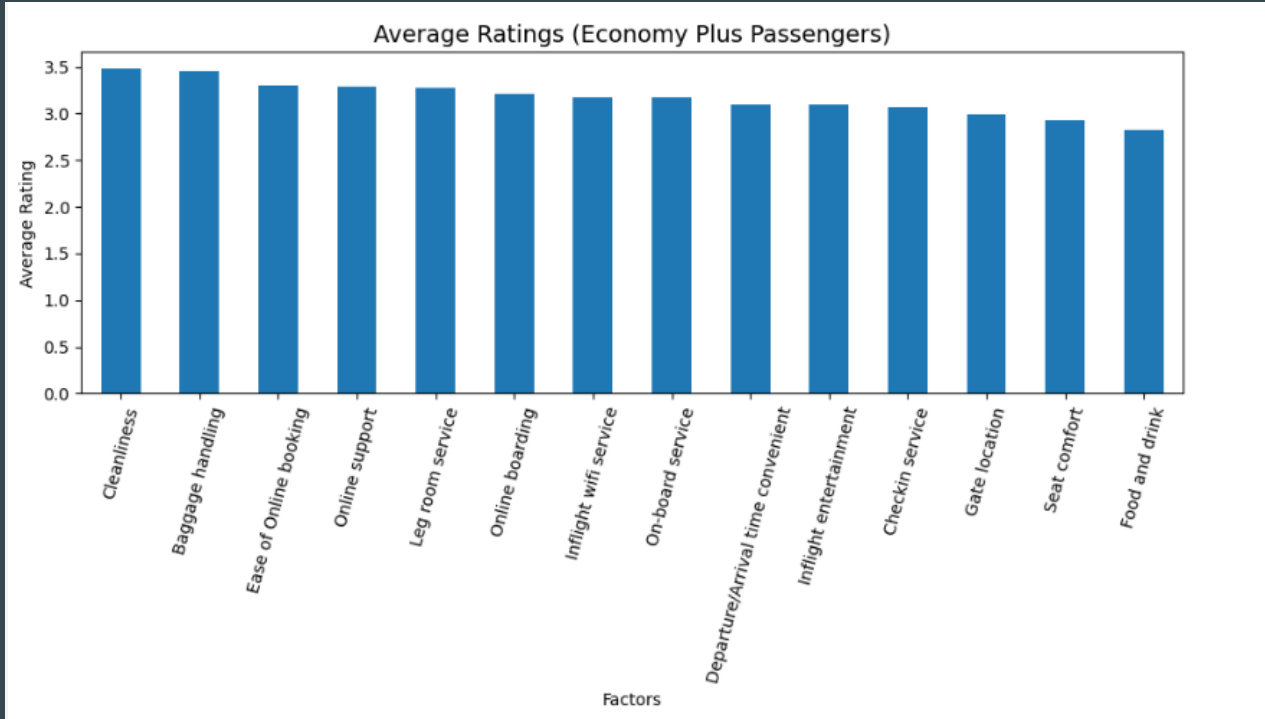
- In average ratings for Business Class we see cleanliness and baggage handling scoring highest, while food & drink and seat comfort have the lowest score.
- From Gate location onwards we see a greater drop in rating

Economy-Plus Factor Correlation Analysis:



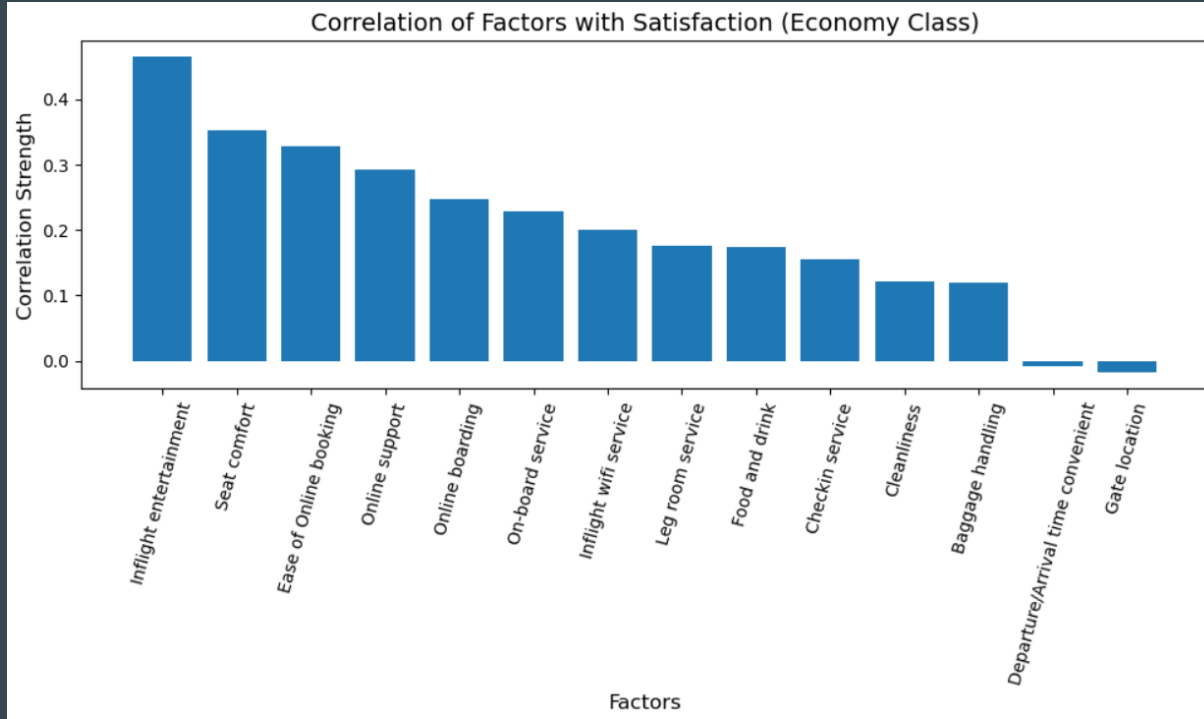
- Inflight entertainment, online booking ease, and seat comfort display the strongest positive correlations.
- Similar to overall correlation, we see Inflight entertainment, online booking ease take the top two spots.

Economy-Plus Average Rating:



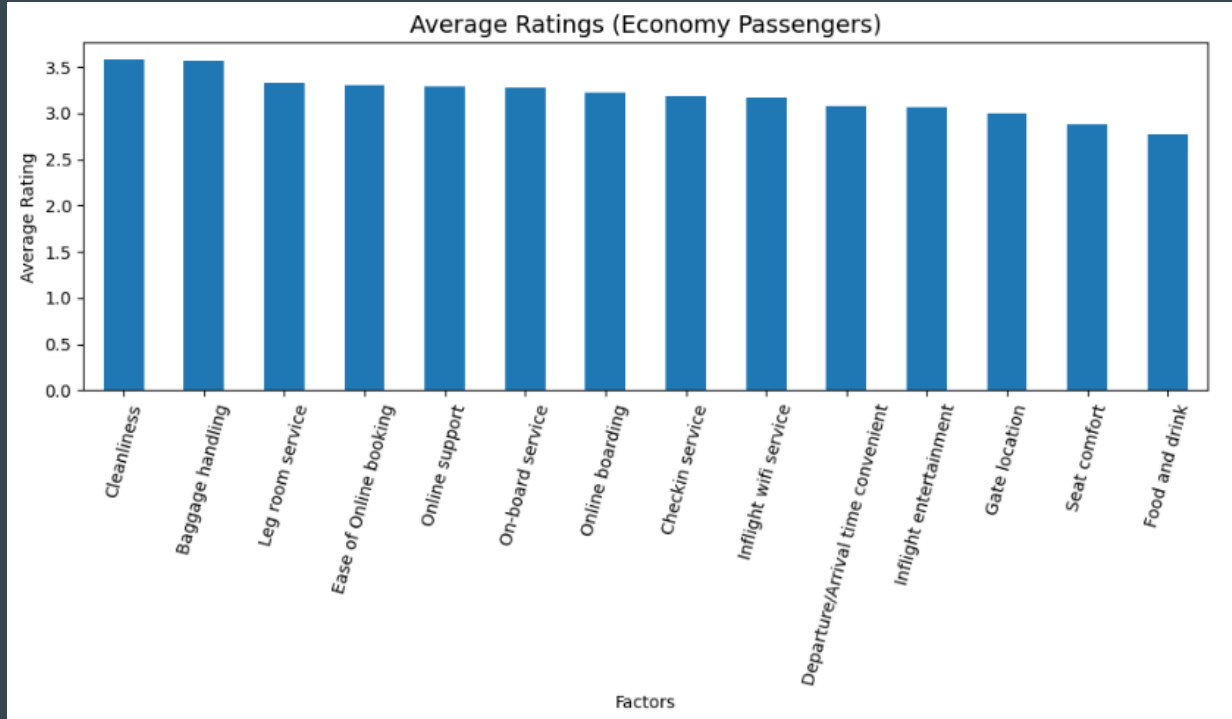
- Cleanliness and baggage handling are rated as the strongest average rating, while food & drink receiving the lowest.

Economy Factor Correlation Analysis:



- This bar chart suggests that Economy and Economy Plus, share the same top three factors.
- Indicating the the two classes share similar values

Economy Factor Average Rating:



- These average ratings from Economy-Class passengers complement the earlier correlation results by showing that factors like cleanliness, baggage handling, and legroom are rated positively overall

Recommendation by Class:

Business Class:

Seat Comfort: Lower-rated compared to other Business Class services — improving comfort enhances the overall premium feel.

Food & Drink: Passengers expect higher quality at business class, so upgrading meals and presentation strengthens satisfaction.

Check-In Service: Enhancing speed, priority handling, and support improves the pre-flight experience for business travelers.

Economy-Plus and Economy Class:

Economy and Economy Plus have similar recommendations:

- **Food & Drink:** Lowest-rated service — biggest opportunity to lift satisfaction.
- **Seat Comfort:** Strong driver of satisfaction with below-average ratings — needs enhancement.
- **Inflight WiFi:** Improves productivity and entertainment — moderate ratings indicate room to upgrade.

Overall Recommendation:

	Correlation_with_Satisfaction	Average_Rating
Inflight entertainment	0.523496	3.383477
Ease of Online booking	0.431772	3.472105
Online support	0.390143	3.519703
On-board service	0.352047	3.465075
Online boarding	0.338147	3.352587
Leg room service	0.304928	3.485902
Checkin service	0.266179	3.340807
Baggage handling	0.260347	3.695673
Cleanliness	0.259330	3.705759
Seat comfort	0.242384	2.838597
Inflight wifi service	0.227062	3.249130
Food and drink	0.120677	2.851994
Gate location	-0.012071	2.990422
Departure/Arrival time convenient	-0.015507	2.990645

- From this table we can derive strong influencing factors that airlines should focus on:
 - Enhancing inflight entertainment
 - Improve online experience
- As well low rated factors that hold some influence on satisfaction that can improve such as:
 - Seat comfort
 - Food and drink

Next Steps:

- Improvements to the charts, more descriptive
 - (colour, bubble chart, heat map)

Machine Learning Model:

- Develop a machine learning model to further analyze our dataset and findings
- These would include but not limited to:
 - Decision-Tree Classifier
 - Sentiment Intensity Analyzer
 - etc.

Conclusion:

- Airlines must balance improvements between high-impact factors and fixing low-rated factors.
- Key Findings:
 - Physical factors such as seat comfort often had low ratings
 - Online factors often fell in the top to top mid range of correlation
- Overall insights
 - No factor scored above a 4 in their ratings, indicating there is a vast room for improvement throughout all factors
 - Economy and Economy plus having similar recommendations highlights that rather than class specific improvements, pattern based improvements are needed

Questions and Answers